Dániel Sajben

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LinkedIn · GitHub · Portfolio

SUMMARY

AI/ML engineer and data scientist who has shipped production LLM systems on AWS and published applied ML research across NLP, graphs and computer vision, with a first-author paper at **IEEE CAMAD 2026**. MSc Advanced AI student at University College Dublin (full-tuition scholarship, graduating Aug 2027) and First Class Honours CS graduate. I like to play poker and calculate the odds, and I am a proud member of the UCD Mathematics Society.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Java, C++, JavaScript, TypeScript, Julia (JuMP)

ML / DL: PyTorch, TensorFlow/Keras, Hugging Face, scikit-learn, CNNs, Transformers, VAEs, diffusion models, GNNs

LLM Systems: RAG, agentic pipelines (LangChain, LangGraph), MCP, fine-tuning, prompting, LLM evaluation, vector databases

Data & MLOps: Apache Spark, Pandas, NumPy, statistics, A/B testing, Matplotlib, Git, Docker, AWS, Linux, Power BI

Web: React, Angular, Spring Boot, REST APIs

Languages: English (C1, IELTS 8.0), German (B1, certified), Hungarian (native)

EDUCATION

University College Dublin — MSc Advanced AI · full-tuition scholarship — Expected *Sep 2026 – Aug 2027*

Technical University of Munich (TUM) — Exchange, CS / Medical AI · GPA 1.3/5.0 (Best: 1.0) — *Oct 2025 – Mar 2026*

Budapest University of Technology and Economics (BME) — BSc CS Eng. · 4.61/5.0, **First Class H.** — *Sep 2022 – Jan 2026*

EXPERIENCE

PTC — Data Science & AI Engineer Intern · (Part-time) — *Mar 2026 – Jun 2026*

- Improved answer relevance on the internal evaluation set by building an agentic **RAG** system in LangGraph and FastMCP with embedding-based HyDE, query rewriting, and decomposition.
- Took an MCP service end to end, from design to production deployment and monitoring on **AWS** (Docker, Linux), integrated with Microsoft 365 Copilot Studio.
- Contributed to customer-facing Codebeamer AI features used by **100K+** users; contract extended for this work.

BME & Ericsson — AI Research Intern · (Part-time) — *Feb 2026 – Aug 2026*

- Reached **92%+** F1 on telecom graph validation, measured with LLM benchmarks and ablation studies, by building tool-grounded **LLM** workflows on locally hosted open-weight models.
- Built a data pipeline of **2K+** synthetic XML graphs for **GNN**/GAN training, presented findings to a global engineering team, and led the AI Summit Budapest 2026 demo.

BME — Research Assistant, Neuroimaging · (Parallel to industry internships) — *Jan 2025 – Aug 2026*

- Won a **Scientific Students' Conference award** (2025) by training a latent diffusion model in **PyTorch** for fMRI super-resolution and data augmentation.
- Cross-validated a VAE + PCA harmonisation pipeline across scanners; submitted to **NeurIPS 2026** as first author.
- Ran a leakage-free statistical comparison against **12** baselines (ComBat, CovBat in Python; RELIEF in **R** via an Rscript bridge) across 4 cohorts; the manuscript now targets the *Medical Image Analysis* journal.

Morgan Stanley — Software Engineer Intern (Institutional Security Trading) · (Full-time) — *Apr 2025 – Aug 2025*

- Shipped production **Java** services and full-stack tools (Spring Boot, Angular) used by **1K+** internal users.
- Decided which UI changes to keep, measured by **A/B tests** on Kibana product telemetry, and communicated results to a cross-functional team.
- Asked by colleagues and managers to demo how I use AI tools (Claude) in my development workflow.
- Cut response times under load by optimising **SQL** queries on financial-data workflows.

SELECTED PROJECTS

Survival Analysis with Tabular Foundation Models — TUM — *Oct 2025 – Feb 2026*

- Reached a C-index of 0.87 (SEER) and 0.884 (PBC), competitive with prior SOTA, by training time-to-event models on embeddings from tabular foundation models.
- Ran that pipeline on 1M+ records by preparing features and cohort joins in **Spark**, then combining survival stacking with in-context learning.
- Presented the results as a poster at a **TUM Medical AI Conference**.

Skin Cancer Classifier — multi-modal computer vision (CNN + RF), BME — *Sep 2024 – Jan 2025*

- Combined **CNN** transfer learning on dermoscopy images with a random forest on patient metadata (380K+ samples).
- Hit **90%** accuracy on the balanced test set after handling class imbalance and applying targeted augmentation; earned the top mark with exam exemption and a strong recommendation letter placing me in the top 10% of students.

PUBLICATIONS & AWARDS

- **Publications**: **IEEE CAMAD 2026** (accepted, first author)
- **Awards:** **Students' Scientific Conference** Award 2025 · **PwC AI Hackathon 2026** Hungary finalist.
- **Anthropic Certifications:** Building with the Claude API · MCP Advanced Topics